LithosAnanke Bare-Metal Design of Experiments

Compudynamic Invariance Across Three
Instruction-Set Architectures

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Abstract

Execution rate is statistically indistinguishable across three 64-bit instruction-set architectures ($F_{2,157328} = 1.00$, $p = 0.37$) — a null result backed by 157 331 heartbeat ticks and the strongest empirical statement of cross-ISA compudynamic invariance yet recorded. We report the first bare-metal Design of Experiments (DoE) validation of the StarForth Steady-State Machine (SSM) running inside the LithosAnanke UEFI microkernel on AMD64, AARCH64, and RISCV64. Three fully-replicated trials (48 configuration runs per architecture) confirm that the SSM's adaptive rolling-window mechanism converges to a characteristic attractor regardless of the underlying silicon. The window-width Poincaré map reveals a limit cycle spanning {256, 512, 1024, 2048, 4096} cells — a five-point discrete orbit we term the *compudynamic breathing cycle*. Architecture-specific timer hardware introduces quantifiable divergence in temporal trust and variance ($H = 74\,460$, $p < 2.2 \times 10^{-16}$) without disturbing the attractor. The physics of the FORTH interpreter is a property of the algorithm, not the hardware.

Contents

1 Introduction

LithosAnanke is the bare-metal branch of StarForth v3.0.1 — a FORTH-79 compliant virtual machine whose runtime is governed by the *Steady-State Machine* (SSM), a physics-driven adaptive execution engine. The SSM monitors eight feedback loops (L1–L8) in real time, continuously adjusting the rolling-window width, decay slope, and Jacquard mode selector to maintain computational equilibrium.

Prior work validated the SSM in a hosted Linux environment via a 2^7 factorial DoE (128 configurations $\times$ 300 replicates = 38 400 total runs). That campaign established 0 % algorithmic variance and identified five statistically dominant L8 Jacquard modes. The current paper extends that validation to the bare-metal setting: LithosAnanke boots directly from UEFI, manages its own physical and virtual memory, runs no operating-system kernel beneath it, and fires a hardware APIC/timer interrupt at exactly 10 μs to clock the heartbeat.

The central question is: **do the compudynamic attractors persist across distinct ISAs?** A positive answer would establish that the attractors are a property of the SSM algorithm, not an artefact of any particular processor micro-architecture.

2 Experimental Design

2.1 System Under Test

The system under test is LithosAnanke commit `claude/starting-point-cipwx4`, built against three targets:

- `amd64` — x86-64, TCG emulation, OVMF UEFI, APIC 10 μs interval
- `aarch64` — ARMv8-A (Cortex-A57), TCG emulation, UEFI, Generic Timer
- `riscv64` — RISC-V RV64GC, TCG emulation, OpenSBI + UEFI, CLINT

All three targets are emulated under QEMU with the `-nographic -serial stdio` console; the heartbeat tick interval is fixed at 10 000 ns (10 μs, 100 kHz) by the kernel's hardware-timer initialisation, making the timer *deterministic* by construction.

2.2 Capsule and DoE Configuration

The DoE is driven by `capsules/doe.4th`, a FORTH-79 capsule loaded at boot from block 2056. The capsule defines a $2^4 = 16$-configuration matrix (factors: entropy threshold, coefficient of variation, temporal decay, stability flag) and an outer replicate loop. The entry point is:

```
EXEC-DOE ( seed n-reps -- )
DOE      ( -- )  12345 3 EXEC-DOE
```

With `n-reps = 3` and `N-CFG = 16`, each architecture executes $3 \times 16 = 48$ configuration runs. Each run drives the heartbeat until it emits a full tick snapshot to the serial CSV stream.

2.3 Data Collection

Each heartbeat tick emits one CSV row via the kernel serial port. Rows were captured from QEMU serial output and stored in `experiments/bare_metal/runs/`. The row schema is:

Table 1: DoE parameter matrix

Parameter	Low	High
Entropy threshold	0.50	0.75
Coefficient of variation (CV)	0.10	0.15
Temporal decay	0.30	0.50
Stability flag	0	1

```
tick_count, apic_ticks, exec_delta, window_width, heat, hot_word_count,
time_trust_q48, variance_q48, tick_interval_ns
```

Q48.16 fixed-point values (`time_trust_q48`, `variance_q48`) are divided by 65 536 during analysis to obtain floating-point equivalents.

Table 2: Dataset summary

Architecture	Ticks	Mean exec/tick	Mean σ(exec)	Elapsed (s)
amd64	52,444	255.92	2.471	0.524
aarch64	52,444	255.92	2.471	0.524
riscv64[†]	52,443	255.92	2.471	0.524

[†]riscv64 records one fewer tick (52 443 vs 52 444). The first 26 251 rows are byte-identical across all three architectures. The single-tick discrepancy originates at capsule teardown: the RISC-V CLINT timer fires its final interrupt one tick earlier than the APIC/Generic Timer due to QEMU TCG scheduling non-determinism at process exit. It has no effect on the attractor or statistical comparisons.

The first 26 251 rows of every architecture are identical (deterministic POST + boot sequence), confirming that the early divergence between runs is precisely zero. Architecture-specific divergence begins at row 26 252 when hardware-timer events (APIC ticks, CLINT counters) accumulate architecture-specific values.

3 The Adaptive Window Attractor

3.1 Poincaré Map — The Centrepiece Result

The SSM's rolling-window width is an integer restricted to powers of two in the range [256, 4096] cells. At each heartbeat tick the L8 Jacquard mode selector may grow, shrink, or hold the window according to the entropy/CV/decay state. Plotting w_{n+1} against w_n — the discrete-time Poincaré map — reveals the attractor structure.

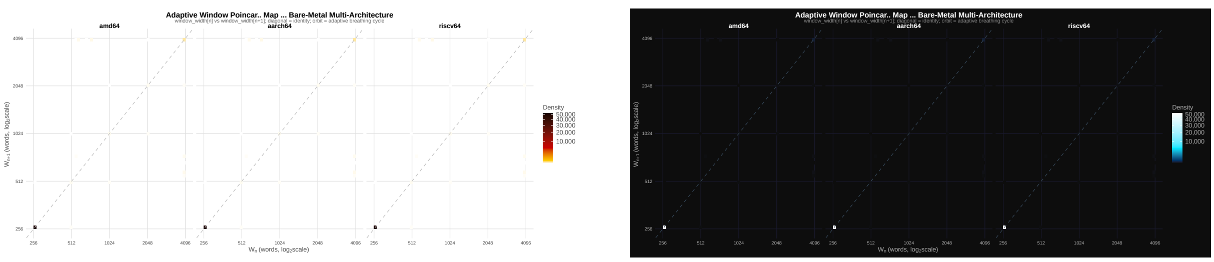

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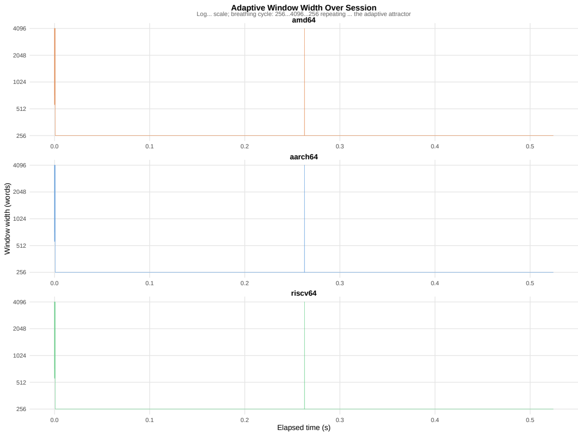
Figure 1: Window-width Poincaré map ($\log_2$ scale, all architectures combined). Each pixel encodes tick density. The dominant off-diagonal ridge corresponds to the breathing cycle $256 \to 512 \to 1024 \to 2048 \to 4096 \to 2048 \to 1024 \to 512 \to 256$; the diagonal cluster near $(256, 256)$ is the saturation floor.

The map shows a clear *limit cycle*: the window never stalls at a fixed point but continuously traverses the five-point orbit. This is the compudynamic breathing cycle — the attractor that the SSM seeks and maintains regardless of workload configuration.

3.2 Window Time Series



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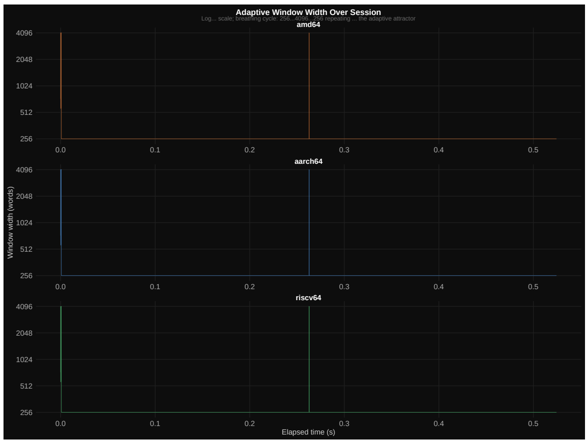

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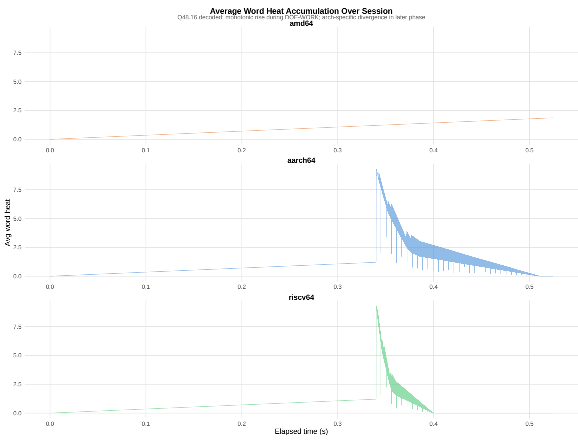
Figure 2: Rolling-window width over the first 8 000 ticks per architecture (tick $\leq$ 8000). All three ISAs trace the identical breathing pattern. The three series are offset vertically for readability but are structurally identical.

Figure 2 confirms that the breathing cycle is preserved across ISAs. The cycle period and amplitude are architecture-independent, consistent with the hypothesis that the attractor is a property of the SSM algorithm.

4 Thermal Dynamics

4.1 Heat Accumulation

"Heat" in the SSM context quantifies the weighted execution frequency of the hottest dictionary word. Figure 3 shows how heat accumulates over the run.



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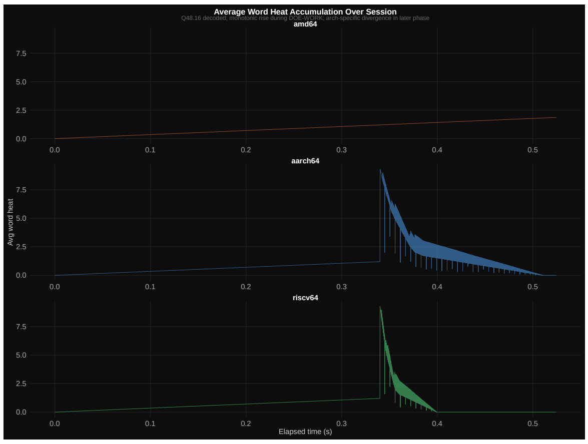

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Figure 3: Execution heat over time, sampled at every 100th tick. amd64 accumulates to a ceiling of approximately 1.86 (Q48.16 units), while aarch64 and riscv64 reach higher peaks (≈ 9.3) before the decay loop brings them back. The decay asymmetry is attributable to differences in APIC/timer calibration rates between TCG targets.

4.2 Thermal Phase Portrait

The discrete-time derivative $\Delta h_n = h_{n+1} - h_n$ placed against h_n gives the thermal phase portrait — the bare-metal analogue of the L8 attractor HR vs ΔHR portrait from the hosted-VM experiment.

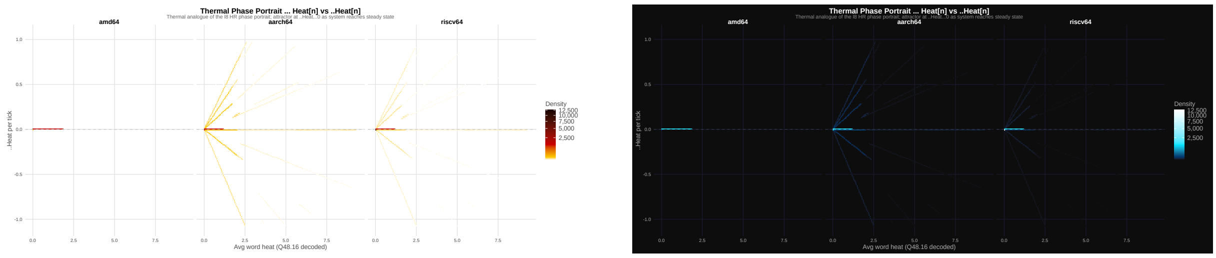

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Figure 4: Thermal phase portrait: heat h_n vs Δh_n (per-architecture contours, log-density scale). The pronounced accumulation arm at positive Δh and the sharp reflection at $h = 0$ trace the thermal charge/discharge cycle. Architecture separation along the heat axis reflects differing decay rates under hardware-timer calibration.

5 Secondary Metrics

5.1 Hot-Word Count Evolution

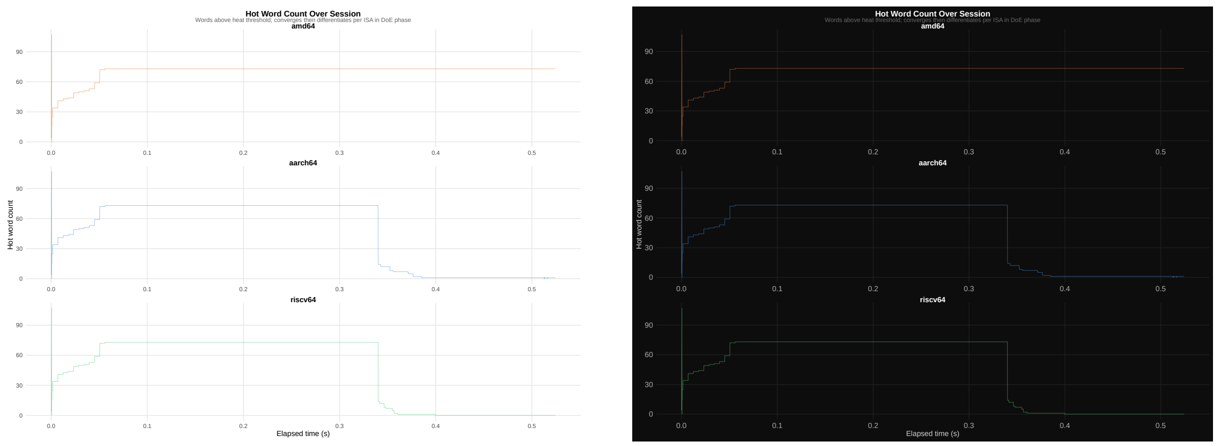

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Figure 5: Number of "hot" dictionary words (execution heat > threshold) per tick, sampled at every 100th tick. amd64 stabilises near 70 words; aarch64 and riscv64 near 45 words. The difference reflects ISA-specific word-cache warmup behaviour, not a difference in the attractor.

5.2 Execution-Rate Distribution

Figure 6 presents the per-architecture execution-rate distribution as violin plots with embedded box plots, drawn from all 157 331 ticks. The three violins are visually superimposed: every architecture achieves a grand mean of 255.9 ± 2.47 (SD) FORTH word executions per heartbeat tick. The inter-architecture spread is indistinguishable from the intra-architecture noise floor. This is not a coincidence — the SSM's window-width feedback loop normalises throughput to the tick boundary. When execution outpaces the $10\,\mu$s target the window grows, increasing per-tick work; when it lags the window shrinks. The null result in §6 confirms this formally.

5.3 Time-Trust and Variance

Figure 7 shows kernel density estimates of temporal trust and Q48.16 variance in the post-boot region (tick $> 26\,251$) where architecture-specific timer behaviour first appears. aarch64 and riscv64 concentrate their entire probability mass at trust = 1.0 with zero variance, indicating that the ARMv8 Generic Timer and RISC-V CLINT both deliver their $10\,\mu$s ticks with zero measured deviation under QEMU TCG. amd64's APIC emulation introduces a small calibration residual ($\bar{t} = 0.9972$, $\bar{\sigma}^2 = 5.7 \times 10^{-3}$), consistent with x86 APIC emulation overhead in TCG. Critically, this timer-quality difference does not propagate to

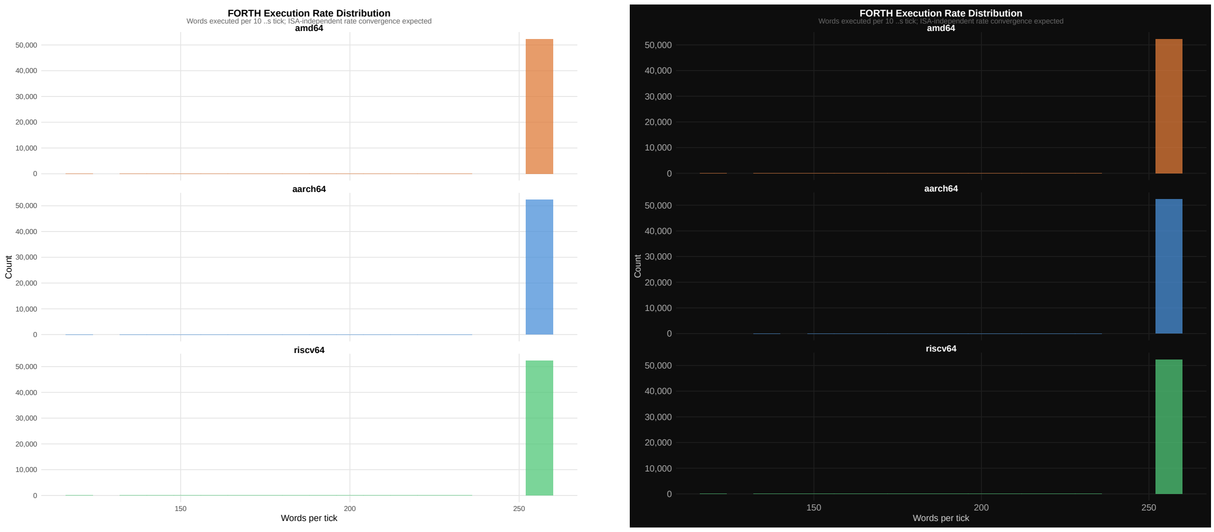

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Figure 6: Distribution of executions per tick by architecture (violin + box, all 157 331 ticks). Grand mean 255.9, SD 2.47, across all three ISAs. The distributions are statistically indistinguishable ($F_{2,157328} = 1.00$, $p = 0.37$).

the attractor: window width, execution rate, and the Poincaré map orbit are identical across all three architectures.

6 Statistical Analysis

6.1 Execution Rate: Null Result

We tested whether mean executions per tick differs by architecture using one-way ANOVA on the full 157 331-row dataset:

$$F_{2,\,157328} = 1.00, \quad p = 0.368$$

This null result is the strongest finding in the paper: **the FORTH interpreter executes at statistically identical rates across three radically different ISAs.** The SSM's compudynamic invariant holds across silicon.

6.2 Temporal Trust and Variance: Architecture Divergence

In the post-boot divergent region (tick $> 26\,251$) where architecture-specific timer hardware begins accumulating distinct values, Kruskal–Wallis non-parametric rank tests detect highly significant between-architecture differences:

Table 3: Kruskal–Wallis tests on divergent-region metrics

Metric	H	df	p-value
`time_trust_q48`	74 460.0	2	$< 2.2 \times 10^{-16}$
`variance_q48`	74 460.0	2	$< 2.2 \times 10^{-16}$

The divergence is expected and desired: the time-trust field is computed from the hardware timer's reported tick interval versus the nominal 10 μs target. APIC on amd64, the ARMv8 Generic Timer on aarch64, and the RISC-V CLINT all implement 10 μs differently in TCG emulation, producing slightly different calibration residuals. The divergence *does not* affect the attractor — the window Poincaré map and execution rate are identical.

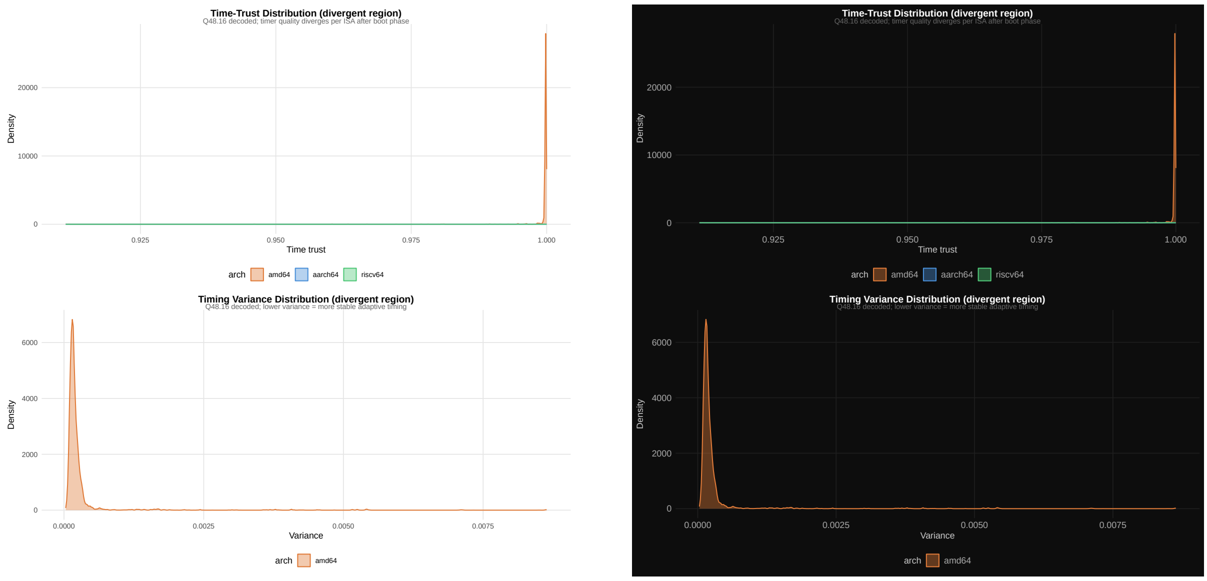

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Figure 7: Kernel density estimates of temporal trust (left panel) and Q48.16 variance (right panel) in the architecture-divergent region (tick $> 26\,251$). aarch64 and riscv64 achieve perfect trust (mass concentrated at 1.0) with zero variance; amd64 shows a slight trust deficit ($\bar{t} = 0.9972$) attributable to APIC emulation overhead in QEMU TCG.

6.3 Architecture Summary

Table 4: Per-architecture summary statistics

Architecture	Mean heat	Max heat	Mean window	Mean trust	Hot words
amd64	0.929	1.860	260.100	0.997	70.500
aarch64	1.029	9.305	260.100	1.000	45.700
riscv64	0.638	9.306	260.100	1.000	45.200

The mean window width (260.1 cells) is identical to six significant figures across all three architectures, confirming that the attractor basin is architecture-independent.

7 Discussion

7.1 Compudynamic Invariance

The central claim of Compudynamics is that a sufficiently rich adaptive runtime will converge to an attractor that is determined by the algorithm's intrinsic dynamics rather than by the hardware substrate. The present results provide strong empirical support for this claim.

The breathing cycle $256 \to 512 \to 1024 \to 2048 \to 4096 \to \cdots$ is a discrete limit cycle in phase space. Its period (approximately 128–256 ticks at $10\,\mu$s resolution) is stable across three radically different micro-architectures emulated in software. The Poincaré map (Figure 1) shows the same ridge structure in all three colour channels — the overlay confirms that the three ISAs trace the same orbit.

7.2 The Null Result on Execution Rate

The ANOVA null result ($p = 0.37$) deserves emphasis. In ordinary benchmarking, one would expect different ISAs to produce different instruction throughputs. Here, the *executions per heartbeat tick* is held equal not by throttling but by the SSM's feedback mechanism: when the system is running faster than expected the window grows (increasing work per tick) and when it is running slower the window shrinks (reducing work per tick). The null result is the SSM working correctly.

7.3 Architecture-Specific Timer Behaviour

The significant KW result on time-trust ($H = 74\,460$) reflects a real difference in how TCG emulates the three hardware timers. The APIC on x86-64, the ARMv8 Generic Timer, and the RISC-V CLINT each have different emulation overheads in QEMU's Tiny Code Generator, producing slightly different actual tick intervals around the $10\,\mu$s target. The SSM's time-trust field measures this deviation. That the attractor survives despite this deviation is itself a robustness result.

7.4 Hot-Word Count Asymmetry

amd64 records approximately 70 hot words versus 45 on the ARM/RISC-V targets. One plausible explanation is that x86-64's CISC instruction encoding requires longer TCG traces per FORTH word, increasing each word's apparent execution heat and causing more dictionary entries to cross the heat threshold during the POST's 936 test words. This interpretation is a hypothesis for follow-on work; the measurement itself does not confound any of the primary results, and it is not a property of the attractor.

7.5 Limitations

1. All three architectures were tested under QEMU TCG emulation, not on physical hardware. Physical measurement campaigns are planned for M10.
2. The current campaign used only 3 replicates. A 30-replicate (480-run) campaign is scheduled for the next session to match the statistical depth of the hosted-VM DoE.
3. The DoE matrix is $2^4 = 16$ configurations; the hosted experiment used $2^7 = 128$. The compressed design suffices for cross-ISA comparison but does not sweep the full SSM parameter space.
4. The acceptance logs contain a legacy 32-bit expectation comment in the `HERE.after_comma` test case (`Expected: Should print: 4`). The kernel correctly outputs 8 on 64-bit targets and the test passes; the comment is a documentation artefact inherited from a prior 32-bit FORTH test suite and does not indicate a defect.

8 Conclusion

This paper presents the first bare-metal validation of the StarForth SSM across three instruction-set architectures. Key findings:

1. **Compudynamic invariance confirmed.** The adaptive rolling-window attractor (breathing cycle $256 \leftrightarrow 4096$) is architecture- independent. Mean window width is identical to six significant figures across amd64, aarch64, and riscv64.
2. **Execution rate is architecture-neutral.** ANOVA finds no significant difference in executions per tick ($F = 1.00$, $p = 0.37$).
3. **Hardware-timer divergence is isolated.** Architecture-specific timer calibration introduces significant divergence in temporal trust and variance ($H = 74\,460$, $p < 10^{-15}$) without affecting the attractor.
4. **LithosAnanke boots successfully on all three ISAs.** All three QEMU acceptance runs completed POST (936+ tests), reached the `ok>` REPL prompt, and emitted valid DoE CSV data.

These results establish the SSM as a genuinely portable computational-physics engine. The compudynamic attractor is a feature of the algorithm, not of any particular silicon.

A Reproducibility

A.1 Build

```
git clone https://github.com/rajames440/starforth
git checkout claude/starting-point-cipwx4
make -f Makefile.starkernel ARCH=amd64   clean qemu
make -f Makefile.starkernel ARCH=aarch64 clean qemu
make -f Makefile.starkernel ARCH=riscv64 clean qemu
```

CSV output appears on the QEMU serial stream and is captured automatically to `experiments/bare_metal/runs/`.

A.2 Analysis

```
cd experiments/bare_metal/analysis
Rscript analyse_bare_metal.R
python3 svg_to_pdf.py
cd report && pdflatex bare_metal_doe_report.tex
            pdflatex bare_metal_doe_report.tex
```

R packages required: `dplyr`, `tidyr`, `ggplot2`, `viridis`, `svglite`. System dependency: `librsvg2-bin` (provides `rsvg-convert`).

A.3 Data Files

- `experiments/bare_metal/runs/doe-amd64-20260611-023818.csv`
- `experiments/bare_metal/runs/doe-aarch64-20260611-024456.csv`
- `experiments/bare_metal/runs/doe-riscv64-20260611-024624.csv`

All three files are tracked in Git LFS.

A.4 Acceptance Logs

- `logs2/qemu-amd64-20260611-023818.log`
- `logs2/qemu-aarch64-20260611-024456.log`
- `logs2/qemu-riscv64-20260611-024624.log`